

ENAE 410 – Project Overview



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Objective

The objective of this course project is to provide students with hands-on experience in the conceptualization, design, manufacturing, and testing of a small RC aircraft.

Project Overview

Students will be tasked with designing a small RC aircraft from a provided kit. The project involves several stages, each of which is critical to the successful completion of the final aircraft. The project will be carried out over the duration of the term and will culminate in a test flight and the submission of a final project report.

The aircraft has the following requirements:

- Must weigh less than 250 grams (249.9 g would be acceptable, 250.1 g would not).
- The aircraft may be hand-launched, launched using a launch rail, or can roll on landing gear.
- The aircraft must fly and complete one circuit of a soccer field (TBD which field).
 - This must be true fixed-wing flight (helicoptering at a high pitch angle is not acceptable).
- The aircraft must be primarily 3D printed (in LW-PLA or PLA).
- The aircraft electronic and motor components must be those provided by the class (no outside motors, etc.).

Project Stages

1. **Conceptual Design:** Determine the sizing and overall performance parameters of the RC aircraft based on the components available in the provided kit. This involves a comprehensive understanding of aerodynamics and material considerations.
2. **Detailed Design and Analysis:** Based on the conceptual design, students will perform a detailed design and analysis of the RC aircraft. This will involve simulations and mathematical modeling to validate the conceptual design.
3. **Manufacturing:** Using the school's 3D printers and other tools, students will manufacture their RC aircraft. They will need to ensure their design can be manufactured within the given resources and constraints.
4. **Test Flight:** Towards the end of the semester, students will complete a test flight of the RC aircraft. They should document the performance of the aircraft, any problems encountered, and any necessary adjustments. Scheduling will be announced during the semester.
5. **Final Report:** Students will compile their designs into a final report, detailing the process they followed, the decisions they made, the challenges they encountered, and the performance of their aircraft. The report is due on the last day of the semester (submitted via D2L).

Design Labs

At the start of the term, working through the “design labs” will guide students through some key design tasks. During these labs, teaching assistants will be available to help students with any problems they encounter.

- **Lab 1 – Conceptual Design and Sizing:** Students will establish a preliminary design for an RC aircraft and determine the sizing parameters.
- **Lab 2 – Aerodynamic Analysis in OpenVSP:** Students will conduct an aerodynamic analysis of the aircraft from Lab 1 using OpenVSP to help validate their conceptual design and support the project.
- **Lab 3 – Stability, Range, and Endurance:** Students will investigate the stability of the reference aircraft from Lab 1 and calculate its projected range and endurance.

Interim Report (10% of course grade) – Due October 18, 2026

By this point, you should have made **significant** headway into your design. An “on paper design” should almost be complete and ready for you to begin testing. In a document, provide details on the status of your design, calculations, and design choices made to date. You should view this report as a rough draft of your final report. The course instructors will provide meaningful feedback that will guide toward a good grade on the final report. Clearly, the design need not be complete, but you should aim to demonstrate that you are on track to finishing by the end of the semester. Things to include could be rough or preliminary analysis results, preliminary sketches or drawings. Outline the steps remaining to complete the design and the report.

There is no strict format for this report. Key elements of the interim report are:

- Highlight the engineering requirements and specifications your design must achieve.
- Describe and justify your preliminary design choices such as type of plane (glider/flying wing/etc.).
- What design parameters did you explore/are you exploring? What is their (anticipated/predicted) effect on performance?
- Include any preliminary calculations and design exploration.
- What are the next steps? What remains to be done? You should be quite far along by this point.

Interim Report Rubric

Section	Description	Points
Engineering requirements and specifications	• Highlights the key engineering requirements • Clearly outlines the specifications the design must achieve • Demonstrates understanding of the constraints and objectives	3
Preliminary design choices and justification	• Describes the preliminary design choices (e.g., type of aircraft configuration) • Provides sound reasoning for the choices made • Discusses alternative designs considered, if applicable	5
Exploration of design parameters	• Connected to, but distinct from, the choices arrived at (previous rubric section) • A variety of design parameters have been explored in support of design choices (airfoil, span, chord, layout, configuration, aspect ratio, etc.) • The implications of the variables and how they connect to the specifications set out (e.g., launch speed) are discussed/explored broadly and thoughtfully	5
Preliminary calculations and analysis	• Includes relevant preliminary calculations that support design decisions • Shows a logical flow from assumptions to results • Provides clear explanations and interpretation of the results	4

Section	Description	Points
Plan for completion	• Outlines the remaining steps to complete the design • Clearly states what needs to be done to finalize the design and report • Provides a realistic timeline or milestones for completion • Sufficient progress has been made such that it is likely the team will fly their plane	3

Total: ____ / 20 pts.

Please Note: Elements of the interim report may be directly reused in the final report.

Final Report (25% of course grade) – Due December 4, 2026

The final report should be a comprehensive document reflecting your understanding and implementation of the project from start to finish. It should include:

- **Introduction:** Briefly describe the objective of your project and the approach you took to achieve it.
- **Design Process:** Detail the design process from conceptual design to the final product. Justify your design choices with relevant data and analysis.
- **Analysis:** Detail the analytical work conducted to validate your design. This should include all the calculations, simulations, and software used.
- **Predicted Performance:** A detailed section predicting the performance of your RC aircraft. Provide quantitative estimates of critical flight parameters, such as range, endurance, stability, etc. The predictions should be based on the content covered in class and justified using your analysis.
- **Manufacturing Process / Structural Design:** Describe the process you followed in manufacturing your RC aircraft, highlighting any challenges encountered and how they were overcome.
- **Test Flight:** Document the test flight, including any problems encountered, adjustments made, and how the performance of the aircraft compared to the predictions.
- **Conclusion:** Summarize your findings and experiences.
- **Appendix (if needed):** Include any additional data, calculations, or information relevant to your project.

Assessment of Final Project

- A detailed rubric is uploaded on D2L.

Resources

- A kit containing all of the primary components for the RC plane will be provided.
- The course TAs will be responsible for printing the aircraft for the course.
- Course Textbook: “Aerodynamics and Aircraft Performance”, 3rd Edition” by Marchmann.
- Course lecture notes and other materials on D2L.
- Students may use Solidworks or Fusion 360 to design their aircraft.
- OpenVSP can be used to conduct aerodynamic analysis.

Good luck!